Chapter 6. The Fourier Transform on Rd

Wave equation, radial functions, Radon transform

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6.1 Preliminaries

Motivation. On $\mathbb{R}^d$ rotations join translations and dilations, and polar coordinates split $\mathbb{R}^d$ into radii and directions.

6.1.1 Symmetries

Analysis on $\mathbb{R}^d$ is shaped by **translations** $x \mapsto x+h$, **dilations** $x \mapsto \delta x$ ($\delta > 0$) and **rotations**: linear R with $R(x) \cdot R(y) = x \cdot y$, i.e. $|Rx| = |x|$, or $R^t = R^{-1}$; $\det R = \pm 1$ (**proper** if $+1$). On $\mathbb{R}$: $\pm I$; on $\mathbb{R}^2 = \mathbb{C}$: $z \mapsto e^{i\varphi} z$, $e^{i\varphi}\bar{z}$; for $d = 3$, Euler angles parametrize proper rotations.

6.1.2 Integration on $\mathbb{R}^d$

A continuous f is **rapidly decreasing** if $\sup_x |x|^k |f(x)| < \infty$ for all k; then $\int_{\mathbb{R}^d} f = \lim_N \int_{Q_N} f$ over cubes $Q_N = \{|x_i| \le N/2\}$ (or balls). $\sup |x|^{d+\epsilon}|f(x)| < \infty$ suffices; **moderate decrease** means $\epsilon = 1$. Then

$$\int f(x+h)\,dx = \int f(x)\,dx, \qquad \delta^d \int f(\delta x)\,dx = \int f(x)\,dx, \qquad \int f(Rx)\,dx = \int f(x)\,dx.$$

Polar coordinates. With $x = r\gamma$, $r \ge 0$, $\gamma \in S^{d-1}$, and surface measure $d\sigma$,

$$\int_{\mathbb{R}^d} f(x)\,dx = \int_{S^{d-1}} \int_0^\infty f(r\gamma)\, r^{d-1}\, dr\, d\sigma(\gamma).$$

For $d = 2$: $\int f = \int_0^{2\pi} \int_0^\infty f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta$; for $d = 3$: $x = (r\sin\theta\cos\varphi, r\sin\theta\sin\varphi, r\cos\theta)$, $d\sigma = \sin\theta\, d\theta\, d\varphi$. S^{d-1} has area $\omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ (from $\int e^{-\pi|x|^2} dx = 1$).

6.2 Elementary theory of the Fourier transform

Motivation. Chapter 5 extends with $x\xi$ replaced by $x \cdot \xi$; new is that the transform commutes with rotations, so radial functions have radial transforms.

Multi-indices: $x^\alpha = x_1^{\alpha_1} \cdots x_d^{\alpha_d}$, $(\frac{\partial}{\partial x})^\alpha$, $|\alpha| = \alpha_1 + \cdots + \alpha_d$. $\mathcal{S}(\mathbb{R}^d)$: smooth f with $\sup_x |x^\alpha (\frac{\partial}{\partial x})^\beta f| < \infty$ for all α, β; $\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i x\cdot\xi} dx$.

Proposition 2.1

For $f \in \mathcal{S}(\mathbb{R}^d)$:
(i) $f(x+h) \to \hat{f}(\xi) e^{2\pi i \xi \cdot h}$;
(ii) $f(x) e^{-2\pi i x\cdot h} \to \hat{f}(\xi + h)$;
(iii) $f(\delta x) \to \delta^{-d} \hat{f}(\delta^{-1}\xi)$, $\delta > 0$;
(iv) $(\frac{\partial}{\partial x})^\alpha f(x) \to (2\pi i \xi)^\alpha \hat{f}(\xi)$;
(v) $(-2\pi i x)^\alpha f(x) \to (\frac{\partial}{\partial \xi})^\alpha \hat{f}(\xi)$;
(vi) $f(Rx) \to \hat{f}(R\xi)$ for every rotation R.

Proof. (i)–(v) as in Proposition 1.2, one variable at a time.

(vi) With $y = Rx$ and $R^{-1}y \cdot \xi = y \cdot R\xi$: $\int f(Rx) e^{-2\pi i x\cdot\xi} dx = \int f(y) e^{-2\pi i y \cdot R\xi} dy = \hat{f}(R\xi)$. ∎

Corollary 2.2

The Fourier transform maps $\mathcal{S}(\mathbb{R}^d)$ to itself.

Proof. By (iv), (v), $\xi^\alpha(\partial_\xi)^\beta \hat{f}$ is the transform of a Schwartz function, hence bounded. ■

Corollary 2.3

The Fourier transform of a radial function ($f(Rx) = f(x)$ for all rotations R) is radial.

Proof. By (vi), $\hat{f}(R\xi) = \hat{f}(\xi)$ for all R. ■

The Gaussian $e^{-\pi|x|^2} = \prod_j e^{-\pi x_j^2}$ is radial and its own transform, so $K_\delta(x) = \delta^{-d/2} e^{-\pi|x|^2/\delta}$ has $\widehat{K_\delta}(\xi) = e^{-\pi\delta|\xi|^2}$ and is a good kernel as $\delta \to 0$.

Theorem 2.4 (Fourier inversion and Plancherel)

If $f \in \mathcal{S}(\mathbb{R}^d)$, then $f(x) = \int_{\mathbb{R}^d} \hat{f}(\xi) e^{2\pi i x\cdot\xi} d\xi$ and $\int |\hat{f}(\xi)|^2 d\xi = \int |f(x)|^2 dx$. In particular $\mathcal{F}$ is a bijection of $\mathcal{S}(\mathbb{R}^d)$.

Proof. As in Chapter 5: the multiplication formula $\int f\hat{g} = \int \hat{f} g$ (Fubini) with $g = e^{-\pi\delta|x|^2}$ and the good kernels K_δ give inversion at 0; translate. For Plancherel use $h = f * f^\flat$, $f^\flat(x) = \overline{f(-x)}$, and $\widehat{f * g} = \hat{f}\hat{g}$. ■

6.3 The wave equation in $\mathbb{R}^d \times \mathbb{R}$

Motivation. Fourier transform solves the wave equation and conserves energy; in $\mathbb{R}^3$ spherical means give Huygens' principle, and $\mathbb{R}^2$ follows by descent.

Cauchy problem. Find $u(x,t)$ with $\triangle u = \frac{\partial^2 u}{\partial t^2}$ (speed $c = 1$), $u(x,0) = f(x)$, $\partial_t u(x,0) = g(x)$, $f, g \in \mathcal{S}(\mathbb{R}^d)$.

6.3.1 Solution in terms of Fourier transforms

In x: $\partial_t^2 \hat{u} = -4\pi^2|\xi|^2 \hat{u}$, so $\hat{u} = \hat{f}(\xi)\cos(2\pi|\xi|t) + \hat{g}(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}$.

Theorem 3.1

A solution of the Cauchy problem for the wave equation is

$$u(x,t) = \int_{\mathbb{R}^d} \left[\hat{f}(\xi)\cos(2\pi|\xi|t) + \hat{g}(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}\right] e^{2\pi i x\cdot\xi} d\xi. \tag{6.1}$$

Proof. $\cos(2\pi|\xi|t)$ and $\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}$ are smooth in ξ (even power series in $|\xi|$) and, with their t-derivatives, polynomially bounded for bounded t. As $\hat{f}, \hat{g}$ decay rapidly, differentiate twice under the integral: each term satisfies $\triangle_x = -4\pi^2|\xi|^2 = \partial_t^2$. At $t = 0$ inversion gives $u = f$, $\partial_t u = g$. ■

The **energy** of u is $E(t) = \int_{\mathbb{R}^d} \left(\left|\frac{\partial u}{\partial t}\right|^2 + \sum_j \left|\frac{\partial u}{\partial x_j}\right|^2\right) dx$.

Theorem 3.2

If u is given by (6.1), then $E(t) = E(0)$ for all t.

Lemma 3.3

For $a, b \in \mathbb{C}$ and $\alpha \in \mathbb{R}$: $|a\cos\alpha + b\sin\alpha|^2 + |-a\sin\alpha + b\cos\alpha|^2 = |a|^2 + |b|^2$.

Proof. $e_1 = (\cos\alpha, \sin\alpha)$, $e_2 = (-\sin\alpha, \cos\alpha)$ form an orthonormal basis of $\mathbb{R}^2 \subseteq \mathbb{C}^2$, and the identity is Pythagoras for the coordinates $(a, b) \cdot e_j$. ∎

Proof of Theorem 3.2. By Plancherel and Proposition 2.1(iv), $\int |\partial_t u|^2 = \int \left|-2\pi|\xi|\hat{f}\sin(2\pi|\xi|t) + \hat{g}\cos(2\pi|\xi|t)\right|^2 d\xi$ and $\sum_j \int |\partial_{x_j} u|^2 = \int \left|2\pi|\xi|\hat{f}\cos(2\pi|\xi|t) + \hat{g}\sin(2\pi|\xi|t)\right|^2 d\xi$. By Lemma 3.3 ($a = 2\pi|\xi|\hat{f}$, $b = \hat{g}$, $\alpha = 2\pi|\xi|t$) the integrands sum to $4\pi^2|\xi|^2|\hat{f}|^2 + |\hat{g}|^2$, free of t. ∎

Uniqueness [Supplement]

For any solution with Schwartz-type decay, integrating by parts as in Theorem 2.3, $E'(t) = 2\,\mathrm{Re}\int \partial_t u\,\overline{(\partial_t^2 u - \triangle u)} = 0$. So zero data give $E \equiv 0$, u constant, hence 0: (6.1) is the only such solution.

6.3.2 The wave equation in $\mathbb{R}^3 \times \mathbb{R}$

The **spherical mean** $M_t(f)(x) = \frac{1}{4\pi}\int_{S^2} f(x - t\gamma)\, d\sigma(\gamma)$ averages f over the sphere of radius t about x.

Lemma 3.4

If $f \in \mathcal{S}(\mathbb{R}^3)$ and t is fixed, then $M_t(f) \in \mathcal{S}(\mathbb{R}^3)$; moreover $M_t(f)$ is C^∞ in t and each t-derivative lies in $\mathcal{S}(\mathbb{R}^3)$.

Proof. For $|x| \geq 2|t|$, $|x - t\gamma| \geq |x|/2$, so $|f(x - t\gamma)| \leq C_N(1 + |x|)^{-N}$. Derivatives in x, t pass under the integral as spherical means of derivatives of f (times polynomials in γ), likewise rapidly decreasing. ∎

Lemma 3.5

$\frac{1}{4\pi}\int_{S^2} e^{-2\pi i\xi\cdot\gamma} d\sigma(\gamma) = \frac{\sin(2\pi|\xi|)}{2\pi|\xi|}$.

Proof. The left side is radial ($d\sigma$ is rotation invariant), so take $\xi = (0, 0, \rho)$:

$$\frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi} e^{-2\pi i\rho\cos\theta}\sin\theta\, d\theta\, d\varphi = \frac{1}{2}\int_{-1}^{1} e^{-2\pi i\rho u} du = \frac{\sin(2\pi\rho)}{2\pi\rho}.$$

∎

Hence, by Fubini,

$$\widehat{M_t(f)}(\xi) = \hat{f}(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|t}. \tag{6.2}$$

Theorem 3.6 (Kirchhoff)

For $d=3$, the solution of the Cauchy problem is $u(x,t)=\frac{\partial}{\partial t}(tM_t(f)(x))+tM_t(g)(x)$.

Proof. By Theorem 3.1, (6.2) and inversion, the solution with data $(0,g)$ is

$$u_1=\int \hat g(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}e^{2\pi ix\cdot\xi}d\xi = t\int \hat g\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|t}e^{2\pi ix\cdot\xi}d\xi = tM_t(g)(x),$$

For data $(f,0)$, $u_2=\int \hat f\cos(2\pi|\xi|t)e^{2\pi ix\cdot\xi}d\xi=\partial_t\Big(t\int \hat f\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|t}e^{2\pi ix\cdot\xi}d\xi\Big)=\partial_t(tM_t(f))$. Superpose. ■

Huygens' principle. In $d=1$, $u(x,t)$ depends on the data on $[x-t,x+t]$ (only the endpoints if $g=0$). In $d=3$ it depends only on the data on the *sphere* $|y-x|=t$: a localized disturbance is felt at distance t exactly at time t and then passes (sharp signals).

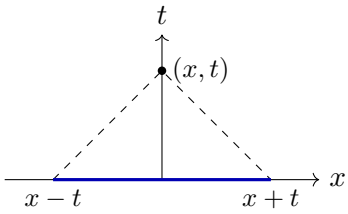


Figure. *Domain of dependence for $d=1$: the interval $[x-t,x+t]$; for $d=3$ only the sphere of radius t.*

6.3.3 The wave equation in $\mathbb{R}^2\times\mathbb{R}$: descent

For $f\in\mathcal{S}(\mathbb{R}^2)$ let $\tilde M_t(f)(x)=\frac{1}{2\pi}\int_{|y|\le1} f(x-ty)(1-|y|^2)^{-1/2}dy$.

Theorem 3.7

A solution of the Cauchy problem in two dimensions with data $f,g\in\mathcal{S}(\mathbb{R}^2)$ is

$$u(x,t)=\frac{\partial}{\partial t}(t\tilde M_t(f)(x))+t\tilde M_t(g)(x). \tag{6.3}$$

Proof.

Step 1 (descent idea). With $\tilde f(x_1,x_2,x_3)=f(x_1,x_2)$, a 3-dimensional solution with data independent of x_3 is independent of x_3 and solves the 2-dimensional problem. Projecting the hemispheres $\gamma=(y,\pm\sqrt{1-|y|^2})$ onto the unit disc, $d\sigma=(1-|y|^2)^{-1/2}dy$ on each, so $M_t(\tilde f)(x,x_3)=\frac{2}{4\pi}\int_{|y|\le1} f(x-ty)(1-|y|^2)^{-1/2}dy=\tilde M_t(f)(x)$.

Step 2 (truncation). As $\tilde f\notin\mathcal{S}(\mathbb{R}^3)$, fix $T>0$, $\eta\in\mathcal{S}(\mathbb{R})$ with $\eta=1$ on $|x_3|\le 3T$, and $\tilde f^\flat=f(x_1,x_2)\eta(x_3)$, $\tilde g^\flat$ likewise. By Theorem 3.6, $u^\flat=\partial_t(tM_t(\tilde f^\flat))+tM_t(\tilde g^\flat)$ solves the 3-dimensional problem. For $|x_3|,|t|\le T$ the spheres stay in $|x_3|\le 2T$, where $\tilde f^\flat=\tilde f$, so $u^\flat$ equals (6.3), independent of x_3; thus $\partial_{x_3}^2u^\flat=0$ and (6.3) solves the 2-dimensional equation for $|t|<T$, T arbitrary. ■

In two dimensions $u(x,t)$ depends on the whole disc of radius t: Huygens' principle fails (ripples linger).

6.4 Radial symmetry and Bessel functions

Motivation. For radial $f=f_0(|x|)$, $\hat f=F_0(|\xi|)$; the relation is elementary for $d=1,3$ and involves J_0 for $d=2$.

- $d = 1$ (radial = even): $F_0(\rho) = \int f(x)e^{-2\pi i x\rho}dx = 2\int_0^\infty \cos(2\pi\rho r) f_0(r)\, dr$.
- $d = 3$: by polar coordinates and Lemma 3.5,

$$F_0(\rho) = \int_0^\infty f_0(r)\Big(\int_{S^2} e^{-2\pi i r\gamma\cdot\xi} d\sigma(\gamma)\Big) r^2 dr$$
$$= \int_0^\infty f_0(r)\frac{2\sin(2\pi\rho r)}{\rho r} r^2 dr = 2\rho^{-1}\int_0^\infty \sin(2\pi\rho r) f_0(r)\, r\, dr.$$

- $d = 2$: the **Bessel function** $J_n(\rho) = \frac{1}{2\pi}\int_0^{2\pi} e^{i\rho\sin\theta}e^{-in\theta}d\theta$ is the n-th Fourier coefficient of $e^{i\rho\sin\theta}$, so $e^{i\rho\sin\theta} = \sum_n J_n(\rho)e^{in\theta}$. With $\xi = (0, -\rho)$,

$$F_0(\rho) = \int_0^{2\pi}\int_0^\infty f_0(r)e^{2\pi i r\rho\sin\theta} r\, dr\, d\theta = 2\pi\int_0^\infty J_0(2\pi r\rho) f_0(r)\, r\, dr.$$

In general F_0 involves a Bessel function of order $d/2 - 1$, elementary only for odd d.

6.5 The Radon transform and some of its applications

Motivation. X-rays measure line or plane integrals of the attenuation; the 1-dimensional transform of the Radon transform is a slice of $\hat{f}$, which allows reconstruction.

6.5.1 The X-ray transform in $\mathbb{R}^2$

A beam of intensity I_0 through an object with attenuation ρ along a line L emerges with $I = I_0 e^{-\int_L \rho}$, so measurements give the **X-ray transform** $X(\rho)(L) = \int_L \rho$. **Reconstruction problem:** recover ρ from $X(\rho)$. In $\mathbb{R}^2$ inversion involves a Hilbert transform; we treat the simpler 3-dimensional analogue.

6.5.2 The Radon transform in $\mathbb{R}^3$

Planes in $\mathbb{R}^3$ are $\mathcal{P}_{t,\gamma} = \{x\cdot\gamma = t\}$, $t\in\mathbb{R}$, $\gamma\in S^2$ ($\mathcal{P}_{t,\gamma} = \mathcal{P}_{-t,-\gamma}$). For $f\in\mathcal{S}(\mathbb{R}^3)$ and $\{e_1, e_2, \gamma\}$ orthonormal,

$$\mathcal{R}(f)(t,\gamma) = \int_{\mathcal{P}_{t,\gamma}} f = \int_{\mathbb{R}^2} f(t\gamma + u_1 e_1 + u_2 e_2)\, du_1\, du_2.$$

Proposition 5.1

For $f\in\mathcal{S}(\mathbb{R}^3)$ and each γ, $\mathcal{R}(f)(t,\gamma)$ does not depend on the choice of e_1, e_2, and $\int_{-\infty}^\infty \mathcal{R}(f)(t,\gamma)\, dt = \int_{\mathbb{R}^3} f(x)\, dx$.

Proof. Another e_1', e_2' differs by a rotation or reflection of $\mathbb{R}^2$, which preserves $\int_{\mathbb{R}^2}$. And $x = t\gamma + u_1 e_1 + u_2 e_2$ is a rotation of $\mathbb{R}^3$, so Fubini gives the identity. ∎

Lemma 5.2

For $f\in\mathcal{S}(\mathbb{R}^3)$ and fixed γ, $\mathcal{R}(f)(\cdot,\gamma)\in\mathcal{S}(\mathbb{R})$, and $\widehat{\mathcal{R}(f)}(s,\gamma) = \hat{f}(s\gamma)$ (one-dimensional transform in t on the left, three-dimensional on the right).

Proof. $|t\gamma + u|^2 = t^2 + |u|^2$ gives $(1+|t|)^N|f(t\gamma+u)| \le C(1+|u|)^{-3}$, so $\mathcal{R}(f)$ decays rapidly in t; t-derivatives are Radon transforms of directional derivatives. Proposition 5.1 for $f(x)e^{-2\pi i s x\cdot\gamma}$ ($= e^{-2\pi i s t} f$ on $\mathcal{P}_{t,\gamma}$) gives $\int \mathcal{R}(f)(t,\gamma)e^{-2\pi i s t}dt = \int_{\mathbb{R}^3} f(x)e^{-2\pi i s x\cdot\gamma}dx = \hat{f}(s\gamma)$. ∎

Corollary 5.3

If $f, g \in \mathcal{S}(\mathbb{R}^3)$ and $\mathcal{R}(f) = \mathcal{R}(g)$, then $f = g$.

Proof. Lemma 5.2 gives $\hat{f}(s\gamma) = \hat{g}(s\gamma)$ for all s, γ, so $\hat{f} = \hat{g}$; invert. ■

The **dual Radon transform** $\mathcal{R}^*(F)(x) = \int_{S^2} F(x \cdot \gamma, \gamma)\, d\sigma(\gamma)$ averages F over the planes through x.

Theorem 5.4 (Radon inversion)

If $f \in \mathcal{S}(\mathbb{R}^3)$, then $\triangle(\mathcal{R}^*\mathcal{R}(f)) = -8\pi^2 f$.

Proof. By Lemma 5.2 and inversion, $\mathcal{R}(f)(t, \gamma) = \int_{\mathbb{R}} \hat{f}(s\gamma) e^{2\pi i t s} ds$, so $\mathcal{R}^*\mathcal{R}(f)(x) = \int_{S^2} \int_{\mathbb{R}} \hat{f}(s\gamma) e^{2\pi i x \cdot \gamma s} ds\, d\sigma(\gamma)$. $\triangle$ multiplies the integrand by $-4\pi^2 s^2$, and $(s, \gamma) \mapsto (-s, -\gamma)$ makes the parts $s < 0$, $s > 0$ equal, so

$$\triangle(\mathcal{R}^*\mathcal{R}(f))(x) = -8\pi^2 \int_{S^2} \int_0^\infty \hat{f}(s\gamma) e^{2\pi i x \cdot s\gamma} s^2 ds\, d\sigma(\gamma) = -8\pi^2 f(x),$$

by inversion in polar coordinates. ■

So $f = -\frac{1}{8\pi^2} \triangle \mathcal{R}^*\mathcal{R}(f)$: plane integrals determine f explicitly, the basis of tomography.

6.6 Selected exercises

Motivation. Rotation invariance, Plancherel with derivatives, and Fubini on $\mathbb{R}^3 \times S^2$ in book exercises.

Exercise 6.5 (Gaussian integrals with a matrix)

Let A be a real symmetric positive definite $d \times d$ matrix. Show that

$$\int_{\mathbb{R}^d} e^{-\pi(x, Ax)}\, dx = (\det A)^{-1/2}.$$

Solution.

Step 1. Take R orthogonal with $R^t A R = D = \operatorname{diag}(\lambda_1, \ldots, \lambda_d)$, $\lambda_i > 0$. With $\lambda = \min_i \lambda_i$, $(x, Ax) \geq \lambda |x|^2$, so $e^{-\pi(x,Ax)}$ is rapidly decreasing.

Step 2. R is a rotation (Section 6.1.1), so $\int f(x)\, dx = \int f(Ry)\, dy$ (Section 6.1.2), and $(Ry, ARy) = (y, Dy) = \sum_i \lambda_i y_i^2$:

$$\int_{\mathbb{R}^d} e^{-\pi(x,Ax)}\, dx = \int_{\mathbb{R}^d} \prod_{i=1}^d e^{-\pi \lambda_i y_i^2}\, dy = \prod_{i=1}^d \int_{\mathbb{R}} e^{-\pi \lambda_i t^2}\, dt,$$

the integral over Q_N factoring before $N \to \infty$.

Step 3. With $s = \sqrt{\lambda_i}\, t$, each factor is $\lambda_i^{-1/2}$, so the integral is $(\det A)^{-1/2}$. ■

Exercise 6.6 (Heisenberg uncertainty principle in $\mathbb{R}^d$)

Let $\psi \in \mathcal{S}(\mathbb{R}^d)$ with $\int |\psi|^2 = 1$. Show that

$$\Big(\int_{\mathbb{R}^d} |x|^2 |\psi(x)|^2 dx\Big)\Big(\int_{\mathbb{R}^d} |\xi|^2 |\hat{\psi}(\xi)|^2 d\xi\Big) \geq \frac{d^2}{16\pi^2}.$$

Solution.

Step 1. Integrating by parts in x_j (other coordinates fixed),

$$\int_{\mathbb{R}} |\psi|^2 \, dx_j = \big[x_j |\psi|^2\big]_{x_j=-\infty}^{\infty} - \int_{\mathbb{R}} x_j \, \partial_j(|\psi|^2) \, dx_j = -\int_{\mathbb{R}} x_j \, \partial_j(|\psi|^2) \, dx_j,$$

the boundary term vanishing by rapid decay; by Fubini, $1 = -\int x_j \partial_j(|\psi|^2)\, dx$.

Step 2. $\partial_j |\psi|^2 = 2\,\mathrm{Re}(\bar{\psi}\, \partial_j \psi)$, so summing over j

$$d = -2\,\mathrm{Re} \int_{\mathbb{R}^d} \bar{\psi}(x) \sum_{j=1}^{d} x_j \, \partial_j \psi(x) \, dx \leq 2 \int_{\mathbb{R}^d} |x| \, |\psi(x)| \, |\nabla \psi(x)| \, dx,$$

as $|\sum_j x_j \partial_j \psi| \leq |x| \, |\nabla \psi|$.

Step 3. By Cauchy–Schwarz for integrals,

$$d^2 \leq 4 \int |x|^2 |\psi|^2 dx \cdot \int |\nabla \psi|^2 dx.$$

Step 4. $\widehat{\partial_j \psi} = 2\pi i \xi_j \hat{\psi}$ and Plancherel (Theorem 2.4) give

$$\int |\nabla \psi|^2 dx = \sum_{j=1}^{d} \int 4\pi^2 \xi_j^2 |\hat{\psi}(\xi)|^2 d\xi = 4\pi^2 \int |\xi|^2 |\hat{\psi}(\xi)|^2 d\xi.$$

so $d^2 \leq 16\pi^2 \int |x|^2 |\psi|^2 \int |\xi|^2 |\hat{\psi}|^2$. ∎

Exercise 6.12 (The dual Radon transform is the adjoint)

For $f \in \mathcal{S}(\mathbb{R}^3)$ and $F \in \mathcal{S}(\mathbb{R} \times S^2)$ prove

$$\int_{\mathbb{R}} \int_{S^2} \mathcal{R}(f)(t,\gamma) \, \overline{F(t,\gamma)} \, d\sigma(\gamma) \, dt = \int_{\mathbb{R}^3} f(x) \, \overline{\mathcal{R}^*(F)(x)} \, dx.$$

Solution.

Step 1. For fixed γ, $h_\gamma(x) = f(x)\overline{F(x \cdot \gamma, \gamma)}$ is continuous, rapidly decreasing, and equals $f(x)\overline{F(t,\gamma)}$ on $\mathcal{P}_{t,\gamma}$, so $\mathcal{R}(h_\gamma)(t,\gamma) = \mathcal{R}(f)(t,\gamma)\overline{F(t,\gamma)}$.

Step 2. The proof of Proposition 5.1 applies to h_γ:

$$\int_{\mathbb{R}} \mathcal{R}(f)(t,\gamma) \overline{F(t,\gamma)} \, dt = \int_{\mathbb{R}^3} f(x) \overline{F(x \cdot \gamma, \gamma)} \, dx.$$

Step 3. Integrate over γ. As $\int_{S^2} \int_{\mathbb{R}^3} |f(x)| \, |F(x \cdot \gamma, \gamma)| \, dx \, d\sigma \leq 4\pi \sup |F| \int |f| < \infty$, Fubini gives

$$\int_{S^2} \int_{\mathbb{R}} \mathcal{R}(f) \overline{F} \, dt \, d\sigma(\gamma) = \int_{\mathbb{R}^3} f(x) \, \overline{\int_{S^2} F(x \cdot \gamma, \gamma) \, d\sigma(\gamma)} \, dx = \int_{\mathbb{R}^3} f \, \overline{\mathcal{R}^*(F)} \, dx.$$

Step 4. On the left, $|\mathcal{R}(f)F| \leq \sup |F| \, \mathcal{R}(|f|)$ and $\int_{\mathbb{R}} \mathcal{R}(|f|)(t,\gamma)\, dt = \int |f|$, so the integrand is integrable on $\mathbb{R} \times S^2$ and the order $\int_{\mathbb{R}} \int_{S^2}$ is allowed: $(\mathcal{R}f, F)_2 = (f, \mathcal{R}^* F)_1$. ∎